

Government Engineering College, Rajkot
3rd Semester - (Odd 2026-27)
Data Structure (BE03000081)

PBL Activity - I

Dead Line : 10/09/2026

Solve following Problems

1. Convert infix expression $a+b*c-d^e+f/g-h$ into postfix expression using stack
2. Evaluate the following Postfix expression using a stack step-by-step. Show the stack content after each token is processed:
6 2 3 * + 4 5 * - 8 2 / +
3. Trace the recursive calls (or stack frames) for the **Tower of Hanoi** problem with $n = 3$ disks and 3 pegs (A, B, C).
4. Consider the following queue, where queue is a circular queue having 6 memory cells.
Front=2, Rear=4 Queue: _, A, C, D, _, _ .
Describe queue as following operation take place:
F is added to the queue
Two letters are deleted
R is added to the queue
S is added to the queue
One letter is deleted
5. Explain tracing of Selection Sort.
42,23,74,11,65,58,94,36,99,87
6. Explain tracing of insertion sort on following data.
42,23,74,11,65,58,94,36,99,87.